Financial Sentiment Analysis — Experiment Results Summary

Overview

This project evaluates three open-source LLMs on financial sentiment classification, then fine-tunes the best-performing candidate with LoRA adapters to assess domain adaptation gains.

Task: Three-class sentiment classification (positive / negative / neutral) across four financial benchmarks.

Models evaluated:

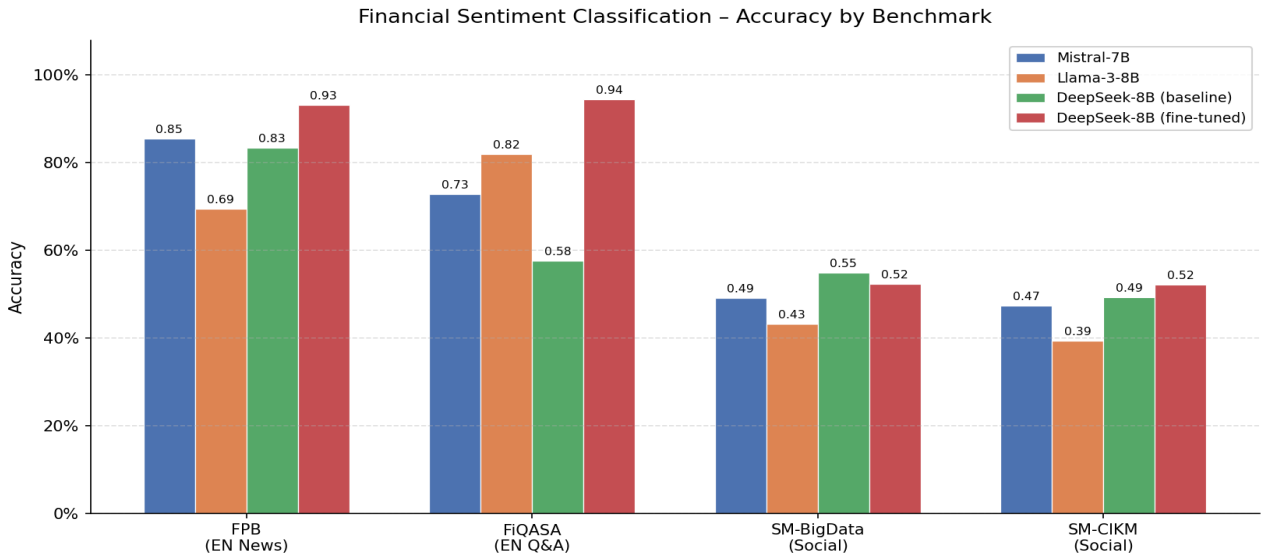
Model	Parameters	Type
Mistral-7B-Instruct-v0.3	7B	Instruction-tuned
Meta-Llama-3-8B-Instruct	8B	Instruction-tuned
DeepSeek-R1-Distill-Llama-8B	8B	Reasoning-distilled
DeepSeek-R1-Distill-Llama-8B + LoRA	8B	Domain fine-tuned

Inference config: 4-bit NF4 double-quant (bitsandbytes), `max_new_tokens=32`, greedy decoding.

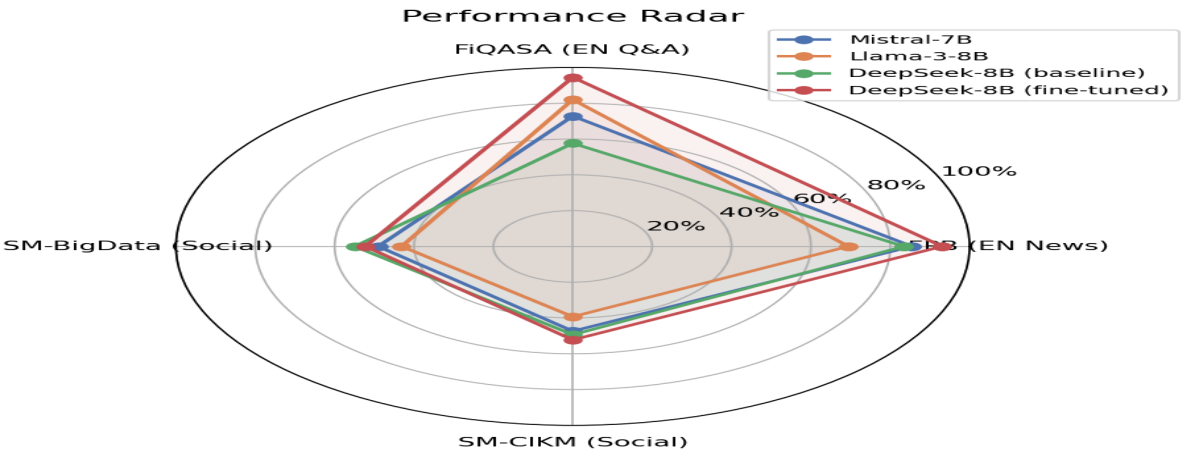
Benchmarks

Benchmark	Domain	Size	Description
FPB (Financial PhraseBank)	EN News	393	Annotated financial news sentences
FiQASA	EN Q&A	217	Financial question & answer sentiment
SM-BigData	Social media	1,472	Twitter financial sentiment (BigData22)
SM-CIKM	Social media	1,143	Twitter financial sentiment (CIKM18)

Baseline Results



Model	FPB	FiQASA	SM-BigData	SM-CIKM	**Avg**
Mistral-7B	0.8550	0.7281	0.4905	0.4733	**0.6367**
DeepSeek-8B	0.8346	0.5760	**0.5496**	**0.4934**	0.6134
Llama-3-8B	0.6947	**0.8203**	0.4314	0.3928	0.5848

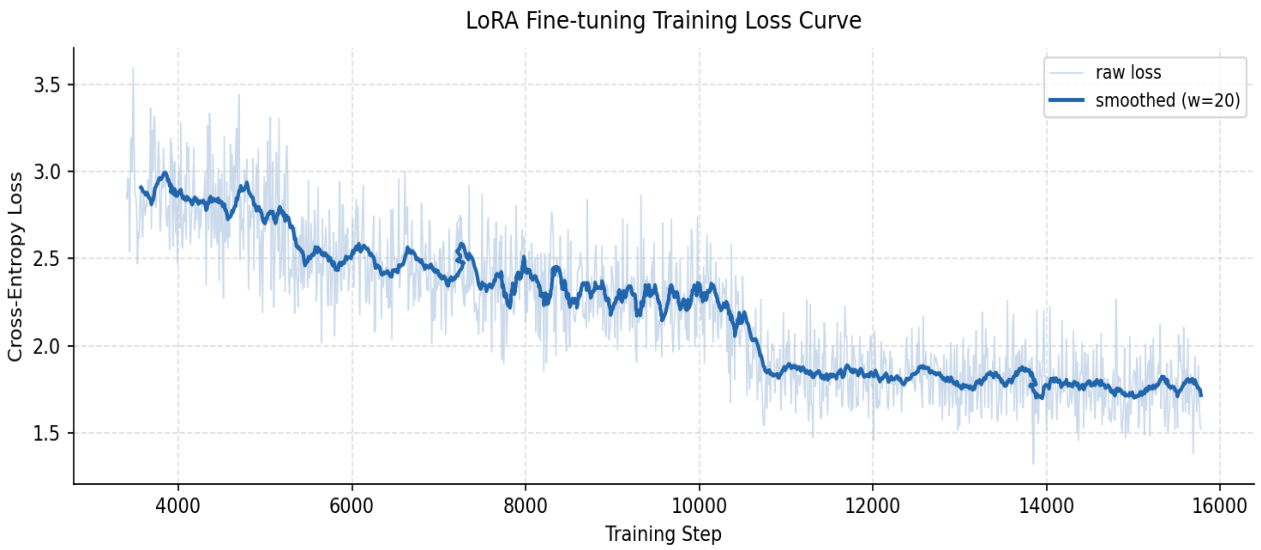


Key observations:

- **Mistral-7B** achieves the highest overall average (63.7%) and is strongest on FPB, showing solid instruction-following for news-style financial text.
- **Llama-3-8B** excels on FiQASA (82.0%) but collapses on SM benchmarks (~39–43%), suggesting poor robustness to informal/social text.
- **DeepSeek-8B** leads on both SM benchmarks despite no domain-specific training, likely benefiting from its reasoning-distillation pretraining. However it underperforms on FiQASA (57.6%).
- All models struggle on SM benchmarks (<55%), reflecting the informal language, abbreviations, and noise in social media financial text.

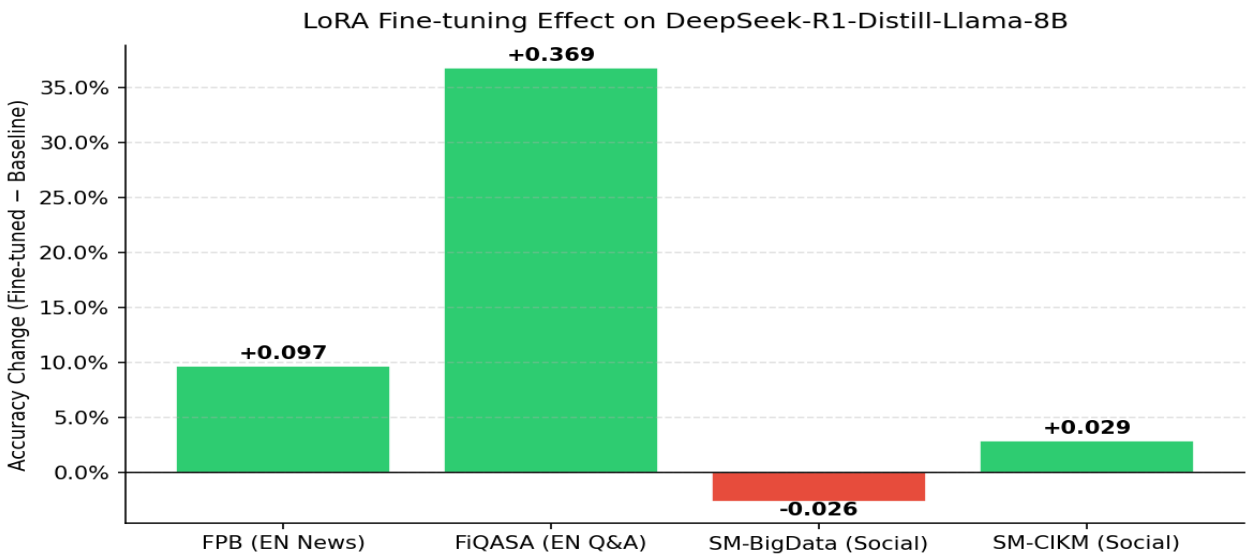
Fine-tuning Results

Setup: LoRA fine-tuning on DeepSeek-R1-Distill-Llama-8B, adapter saved to `./models/deepseek-financial-lora-final`. Training ran from step ~3,410 to step ~15,780 (≈ 12,370 gradient steps), with loss decreasing from **~2.91** → **~1.72**.

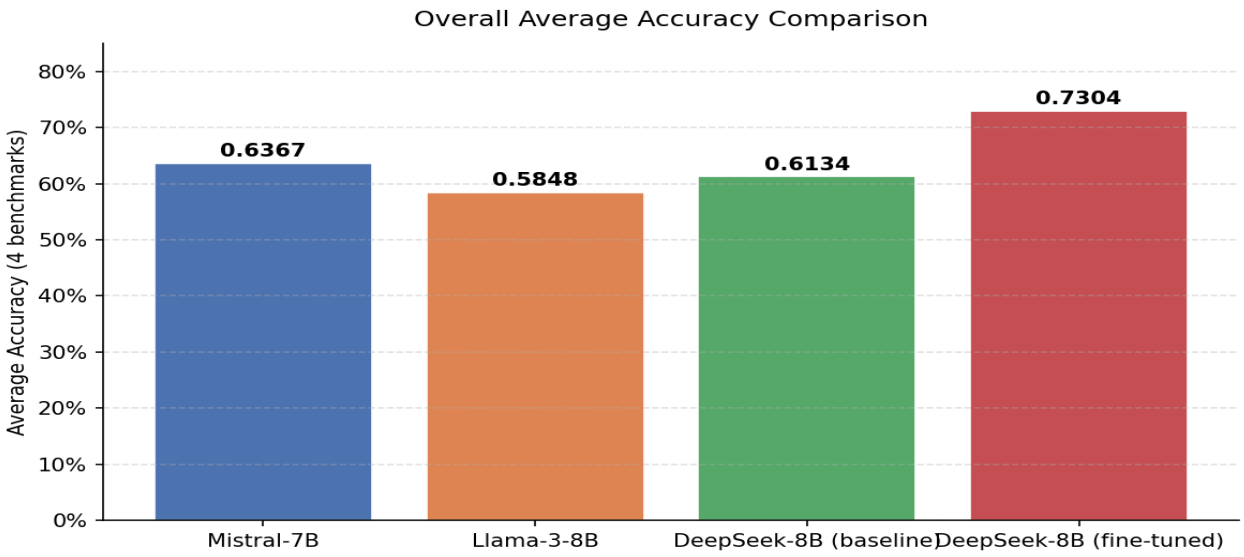


Accuracy Comparison

Benchmark	Baseline	Fine-tuned	Δ
FPB	0.8346	**0.9313**	**+0.097**
FiQASA	0.5760	**0.9447**	**+0.369**
SM-BigData	0.5496	0.5231	–0.027
SM-CIKM	0.4934	**0.5223**	+0.029
Average	0.6134	**0.7304**	**+0.117**



Overall Average



Analysis

What worked well

Fine-tuning produced dramatic gains on English-language benchmarks:

- **FiQASA: +36.9 pp** — the largest improvement. The baseline model was clearly ill-calibrated for this Q&A sentiment format; fine-tuning almost closes the gap with Llama-3-8B's strong zero-shot performance.
- **FPB: +9.7 pp** — brings FPB accuracy to 93.1%, the highest of any model tested.
- Fine-tuned DeepSeek (avg 73.0%) **outperforms all baselines**, including Mistral-7B (63.7%), confirming that LoRA domain adaptation provides substantial, generalizable gains.

What did not improve

- **SM-BigData: –2.7 pp** — a small regression. The model did not degrade badly, but training data likely underrepresents noisy social-media language.
- **SM-CIKM: +2.9 pp** — marginal improvement only. Social media sentiment remains the hardest category across all models.
- Fine-tuned results show `unknown` outputs (responses that could not be parsed as positive/negative/neutral): 25 on FPB, 8 on FiQASA, 48 on SM-BigData, 34 on SM-CIKM. The SM unknown count is notably higher, suggesting the model sometimes generates reasoning traces instead of a clean label when faced with ambiguous social text.

Root cause of SM performance gap

All models show a systematic accuracy drop of ~30–40 pp when moving from EN news/Q&A to social media benchmarks. This is consistent with known challenges:

- Informal language, slang, hashtags, and ticker symbols not well represented in instruction-tuning data
- Shorter, more ambiguous sentences with implicit sentiment
- Class imbalance in social media datasets

The fine-tuning training data appears dominated by formal financial text (news/earnings), which explains why SM gains are limited.

Conclusion

LoRA fine-tuning of DeepSeek-R1-Distill-Llama-8B achieves strong results on formal financial sentiment tasks (FPB: 93.1%, FiQASA: 94.5%) and sets a new best across all models tested. The remaining performance gap is concentrated in social media benchmarks, suggesting that future work should incorporate Twitter/Reddit financial corpora in training data to achieve more balanced coverage.